

No-slip boundary condition in finite-size dissipative particle dynamics

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ABSTRACT

Dissipative particle dynamics (DPD) is an efficient, particle based mesoscopic numerical scheme to simulate dynamics of complex fluids and micro-flows, with spatio-temporal scales in the range of micrometers and microseconds. While the traditional DPD method treated particles as point masses, a modified DPD scheme was introduced recently [W. Pan, I.V. Pivkin, G.E. Karniadakis, Single-particle hydrodynamics in DPD: a new formulation, *Europhysics Letters* 84 (2008) 10012] by including transverse forces between finite sized particles in addition to the central forces of the standard DPD.

The capability of a DPD scheme to solve confined wall bounded flows, depends on its ability to model the flow boundaries and effectively impose the classical no-slip boundary condition. Previous simulations with the modified DPD scheme used boundary conditions from the traditional DPD schemes, resorting to the velocity reversal of re-inserted particles which cross the solid wall. In the present work, a new method is proposed to impose no-slip or tunable slip boundary condition by controlling the non-central dissipative components in the modified DPD scheme. The solid wall is modeled in such a way that the fluid particles feel the presence of a continuous wall rather than a few discrete frozen particles as in conventional wall models. The fluid particles interact with the walls using a modified central repulsive potential to reduce the spurious density fluctuations. Several different benchmark problems (Poiseuille flow, lid-driven cavity and flow past circular cylinder) were solved using the new approach to demonstrate its validity.

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1. Introduction

Numerical simulation of complex-fluid flow through micro and nano-channels has great importance in the field of applied science and research. The characteristic size of these systems is small and hence the Knudsen number is large, making their continuum description questionable [1,2]. Thus modeling of micro and nanoflows has been attempted through microscale discrete models. Several numerical schemes have been developed in the recent past treating the fluid as a collection of discrete particles moving in a Lagrangian fashion obeying laws of classical mechanics. The current repository of such methods include molecular dynamics (MD), dissipative particle dynamics (DPD), stochastic rotation dynamics (SRD), smoothed particle hydrodynamics (SPH), etc. Of these, DPD is a mesoscopic model which lies in between the microscopic MD method and macroscopic computational fluid dynamics (CFD) method.

The DPD model was introduced by Hoogerbrugge and Koelman [3] as a mesoscopic scheme to solve the hydrodynamics of suspensions. Each particle in DPD scheme can be viewed as an agglomeration of molecules and so the relevant spatio-temporal scales are raised from nanoseconds and nanometers to microseconds and micrometers respectively. Subsequently

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this model has proved to be useful for solving problems in the micro- as well as macro- regimes. The dynamics of a variety of complex fluids like suspensions, colloids, polymeric liquids, biological tissues and living cells have been simulated efficiently by adopting this model [2–6]. Classical benchmark problems such as Couette flow, Poiseuille flow, flow past cylinder and spheres, lid driven cavity, oscillating flat plate, annular flows, non-Newtonian fluid flows were solved using DPD and have been shown to compare well with CFD [7–10].

In the standard DPD method [11] the particles are considered to be point masses interacting through three central forces; a conservative soft pure repulsion force, a velocity dependent frictional force and a Brownian stochastic force. In order to treat flow of colloids or fluids containing second phase particles, the larger second phase particles are modeled as an agglomeration of several DPD particles joined using an attractive potential. Since this imposes an additional computational cost, alternative models have been proposed such as the fluid particle model (FPM) [12]. In this model, the particles are considered to be of finite size and both angular and linear momentum balance equations are solved simultaneously to update the position as well as linear and angular velocities of particles. Subsequently, Filipovic et al. [13] included lateral dissipative and random forces in addition to the central forces to simulate flow between concentric cylinders and is termed as transverse DPD (TDPD). Independently, a variant of DPD formulation was introduced by Pan et al. [14] which is capable of handling particles with different sizes by accounting for non-central drag and rotational effects. We use the name finite-size dissipative particle dynamics (FDPD) throughout this paper to distinguish this scheme from other DPD variants.

Numerical simulation of flow through miniaturized devices like ‘lab-on-chips’ or microfluidic devices is crucial for their design and development. These wall bounded flows take place inside a definite flow boundary rather than as part of an infinite domain. To study the flow through these finite geometry problems, the interaction of the fluid with solid surface should be accurately modeled. The solid walls should satisfy *impenetrability* and *no-slip* without causing *temperature and number density fluctuations*. The solid wall has been modeled as a

1. virtual boundary (VB): by modifying periodic boundary conditions without modeling the physical boundaries [15,16],
2. frozen wall boundary (FWB): by keeping layers of interacting but stationary particles (frozen particles) along the shape of boundary outside the fluid domain [2,7,8,10,17–19],
3. sharp wall boundary (SWB): by fixing particles exactly on the physical boundary [20,21].

For geometries simple enough to apply periodic boundary conditions such as Couette flow, the VB method (e.g. Lees–Edwards scheme [15]) has been employed. In this method particles escaping from one boundary reappears from the opposite boundary with modified velocities and positions. This scheme cannot be employed for handling complex geometries and therefore is no longer widely used.

The FWB model is one of the most widely used and several simulations of wall bounded flows have been done by positioning single or multiple layers of frozen particles in a lattice or in random order outside the fluid region. Multiply connected complex bluff geometries like square or circular cylinder in flow can efficiently be modeled by placing the frozen particles in the shape of that boundary. A modified version of FWB known as force boundary condition (FBC) [7] was introduced, by freezing several wall layers combined with bounce back reflection having a new wall particle repulsion parameter which is a function of densities of both wall and fluid.

The third method, the SWB model, requires a fixed layer of immobile particles exactly on the boundary. Different implementations of this scheme with varying maximum conservative repulsion parameter were reported [20,21]. The major difference between FWB and SWB schemes is that, in the latter method only one layer of frozen particles are placed on the boundary, whereas in the former scheme one or more layers extend outside the fluid regime.

Wall impenetrability is mimicked by returning the particles traversing through the solid boundary back into the fluid domain and no-slip is obtained in all the methods discussed so far is by reversing the directions of velocity [21]. However, this causes particle agglomeration near the wall and distortions of the fluid properties are common in this type of implementation.

In this paper we consider the modified DPD model – the finite-size DPD model – and propose appropriate boundary conditions. We model impenetrable solid surfaces without fixing the wall particle prior to simulation. No-slip (or tunable slip) is achieved by adjusting the lateral dissipation coefficient without reverting the directions of velocities of particles. This allows our method to be useful for more complex wall geometries and minimizes near-wall density and temperature fluctuations.

In the next section, a brief description of the governing equations for FDPD are presented. In Section 3 we present details about the new boundary condition. In Section 4 we apply the new method to some classical benchmark problems such as Poiseuille flow, flow in a lid-driven cavity and flow past a circular cylinder and compare results obtained to classical analytical, experimental and numerical results from the literature. Our model does extremely well in achieving no-slip and impenetrability while reducing spurious near-wall number density and temperature fluctuations. The results of our simulations of benchmark problems compare to those from literature to good accord.

2. Finite-size dissipative particle dynamics: theory

In this section a brief description of FDPD model is presented following Pan et al. [14] closely. The FDPD model consists of a system of N particles of finite size characterized by the mass m_i and mass moment of inertia I_i of the i th particle. Each

particle is assumed to be an agglomeration of a large number of molecules and thus the particles interact through conservative inter-particle forces as well as non-conservative forces. The motion of the particles is governed by Newton's linear momentum balance equations

$$m_i \frac{d\mathbf{v}_i}{dt} = \mathbf{f}_i, \quad (1)$$

and the angular momentum balance

$$I_i \frac{d\boldsymbol{\omega}_i}{dt} = - \sum_{j \neq i} \lambda_{ij} \mathbf{r}_{ij} \times \mathbf{f}_{ij}, \quad (2)$$

where $\mathbf{v}_i$ is the linear velocity, $\boldsymbol{\omega}_i$ the angular velocity, $\mathbf{f}_i$ is the force on each particle and $\mathbf{f}_{ij}$ is the total force on the i th particle due to a neighboring j th particle. The position vector of the i th particle with respect to the j th particle is $\mathbf{r}_{ij} = \mathbf{r}_i - \mathbf{r}_j$ with magnitude $r_{ij} = |\mathbf{r}_{ij}|$ and unit vector is given by $\hat{\mathbf{e}}_{ij} = \mathbf{r}_{ij}/r_{ij}$. In this formulation, the tangential force between two particles acts to impart torques on the particles in the ratio of their radii and thus $\lambda_{ij} = \frac{R_i}{R_i + R_j}$ where R_i and R_j are the radii of the i th and j th particles respectively as shown in Fig. 1.

As mentioned earlier, in DPD, each particle is assumed to be a cluster of several molecules and hence the inter-particle forces comprise of a conservative component which may be viewed as the mean effect of the inter-atomic forces, a non-conservative component arising from the dissipation between particles (arising from the dispersion of energy among the large number of constituent molecules) and stochastic components. The total force exerted on i th particle by all its neighboring particles can be written as

$$\tilde{\mathbf{f}}_i = \sum_{j \neq i} \mathbf{f}_{ij} = \sum_{j \neq i} (\mathbf{f}_{ij}^{\text{Cons}} + \mathbf{f}_{ij}^{\text{Diss}} + \mathbf{f}_{ij}^{\text{Rand}}). \quad (3)$$

In addition, an external force may act on each particle and thus the total force is given by

$$\mathbf{f}_i = \tilde{\mathbf{f}}_i + \mathbf{f}_i^{\text{External}}. \quad (4)$$

The conservative force is generally assumed to act along the line of centers of the particles and is taken to be

$$\mathbf{f}_{ij}^{\text{Cons}} = a_{ij} \Gamma(r_{ij}) \hat{\mathbf{e}}_{ij}, \quad (5)$$

where a_{ij} is the repulsion parameter and $\Gamma(r_{ij})$ is an appropriate weight function of the inter-particle distance which has the property that it is repulsive at small separation distances and vanishes for large distances. A commonly used form for $\Gamma(r_{ij})$ in most DPD simulations is

$$\Gamma(r_{ij}) = \begin{cases} 1 - \frac{r_{ij}}{r_c}, & \text{if } r_{ij} < r_c, \\ 0 & \text{if } r_{ij} > r_c, \end{cases} \quad (6)$$

where r_c is the radius of influence sphere.

The nonconservative dissipative forces are assumed to consist of two components (translational and rotational)

$$\mathbf{f}_{ij}^{\text{Diss}} = \mathbf{f}_{ij}^{\text{Trans}} + \mathbf{f}_{ij}^{\text{Rot}} \quad (7)$$

which are taken to be specifically,

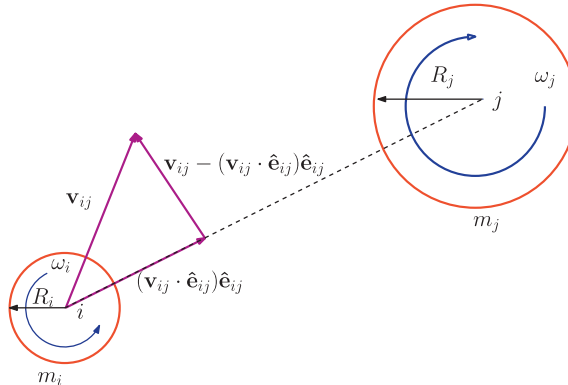


Fig. 1. Schematic representation of the FPD particles. The particles are allowed translational and rotational degrees of freedom.

$$\mathbf{f}_{ij}^{Trans} = -\gamma_{ij}^C \Gamma^2(r_{ij})(\mathbf{v}_{ij} \cdot \hat{\mathbf{e}}_{ij})\hat{\mathbf{e}}_{ij} - \gamma_{ij}^S \Gamma^2(r_{ij})[\mathbf{v}_{ij} - (\mathbf{v}_{ij} \cdot \hat{\mathbf{e}}_{ij})\hat{\mathbf{e}}_{ij}], \quad (8)$$

where $\mathbf{v}_{ij} = \mathbf{v}_i - \mathbf{v}_j$ is the relative velocity and γ_{ij}^C (central), γ_{ij}^S (non-central i.e. shear) are the dissipation coefficients, and

$$\mathbf{f}_{ij}^{Rot} = -\gamma_{ij}^S \Gamma^2(r_{ij})[\mathbf{r}_{ij} \times (\lambda_{ij}\boldsymbol{\omega}_i + \lambda_{ji}\boldsymbol{\omega}_j)]. \quad (9)$$

The stochastic force is given by

$$\mathbf{f}_{ij}^{Rand} \Delta t = \Gamma(r_{ij}) \left[\sigma_{ij}^C \text{tr}[d\mathbf{W}_{ij}] \frac{1}{\sqrt{d}} \mathbf{1} + \sqrt{2} \sigma_{ij}^S d\mathbf{W}_{ij}^A \right] \cdot \hat{\mathbf{e}}_{ij}, \quad (10)$$

where $\text{tr}[d\mathbf{W}_{ij}]$ is the trace of symmetric independent Wiener increment matrix $d\mathbf{W}_{ij}$ (a Gaussian random number matrix with a mean zero and variance Δt , where Δt is the time step) whose antisymmetric counterpart is $d\mathbf{W}_{ij}^A$ and d is the dimensionality of the system, for a two dimensional domain $d = 2$. We note that random and dissipation coefficients are related by $\sigma_{ij}^C = \sqrt{2k_B T \gamma_{ij}^C}$ and $\sigma_{ij}^S = \sqrt{2k_B T \gamma_{ij}^S}$.

3. Instantaneous wall boundary (IWB) method

In this section we will describe the implementation of a novel solid wall boundary condition for the particles governed by the above equations of motion. Fedosov et al. [8] reported that the *wall-normal* conservative force component primarily controls the density perturbations and *wall-tangential* component of dissipative force imposes no-slip near the solid boundary. In the present work we have used the former to suppress the density perturbations near the wall by modifying the particle-wall central repulsion force and the latter to impose the no-slip boundary condition by increasing dissipative force component tangential to the wall.

In FWB method the fluid particles near the wall within cut-off radius r_c interact with a few fixed solid particles depending on the wall number density. Modeling the SWB is also done by placing immovable frozen particles on the wall prior to the computation with a predetermined density. These methods of wall modeling have a few disadvantages. First, the fluid particles interact with discrete fixed particles, rather than a continuous wall. To model a continuous wall, each fluid particle near the wall should feel the presence of wall (through inter-particle forces) at the shortest distance. This is possible with fixed wall particles only if the wall density, $\rho_w = \infty$. At high wall particle number density, the fluid density fluctuations become significant due to the wall effect. Since, due to practical considerations, a finite number of wall particles are used in FWB and SWB, fluid particles penetrate through the wall at a high rate. These particles are reintroduced into the simulation domain by means of specular, bounce forward, bounce backward or Maxwellian reflections [7] and this is a main cause of material property variations near the wall [22]. If the modeled wall ensures *natural* impenetrability without the aid of the artifacts mentioned earlier, this can help control the near-wall fluctuations in fluid properties. The extension of the simulation domain beyond the fluid region in FWB increases the computational cost due to large number of fluid–solid interactions. In addition, the above methods are not easily amenable to tuning the slip-length if required (e.g. for superhydrophobic surfaces [23]). We note that, in continuum based CFD simulations the computational domain is confined within the fluid region, and the wall is not modeled explicitly.

We propose a model for the solid boundary to circumvent the above disadvantages of frozen particle walls. In the proposed method, we do not fix wall particles in space a priori. Instead, the wall interacts with each fluid particle within the cut-off radius as if there were a particle located at the shortest distance from the fluid particle as shown schematically in Fig. 2. In other words, a particle having spatial coordinates (x_p, y_p) which comes within cut-off radius from a planar wall ($y_w = \text{constant}$) may be viewed as interacting with a wall particle at (x_p, y_w) through particle–wall conservative, dissipative and random forces. These “instantaneous frozen particles (IFP)” appear exactly on the boundary dynamically at each time step and disappear after their interaction with the fluid particles is calculated. Thus the wall does not extend beyond the

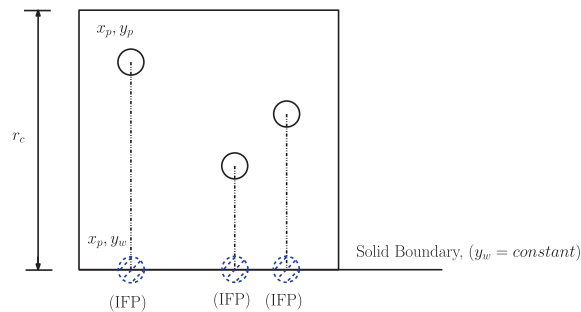


Fig. 2. Each fluid particle within r_c from the solid wall interacts with its corresponding IFP. The size of IFP is that of the fluid particle, its angular velocity is zero and linear velocity is that of solid wall.

fluid domain in this method, and is similar to the SWB method in this aspect. In effect, these virtual wall particles mimic a continuous solid boundary and the wall number density is thus effectively $\rho_w = \infty$.

The main difference between the proposed model and FWB or SWB is that each fluid particle interacts with only one IFP, while in other methods each particle interacts with all the frozen particles which lie within cut-off radius, see Fig. 3. This scheme has the added advantage of better control on the fluid–solid conservative force and, consequently, we are able to efficiently suppress density fluctuations near the wall. The simulation domain is confined within the fluid region so the number of wall particles to be considered is effectively smaller and this allows the method to be computationally more efficient as well.

The conservative interaction force between a fluid particle and the solid wall (henceforth we will use subscript ‘p’ to refer a fluid particle and ‘w’ to refer a solid wall) is taken to be

$$\mathbf{f}_{pw}^{\text{Cons}} = a_{pw} \Psi(r_{pw}) \hat{\mathbf{e}}_{pw}, \quad (11)$$

where a_{pw} is the fluid–solid central repulsion parameter, $r_{pw} = |\mathbf{r}_{pw}|$ and unit vector is $\hat{\mathbf{e}}_{pw} = \mathbf{r}_{pw}/r_{pw}$. In order to control density fluctuations near the wall, the function $\Psi(r_{pw})$ is required to be stiff near the walls and softer away from the wall. An appropriate function which satisfies these requirements is chosen as

$$\Psi(r_{pw}) = \begin{cases} \kappa \left(\frac{1 - (r_{pw}/r_c)^3}{\kappa + (r_{pw}/r_c)^3} \right), & \text{if } r_{pw} < r_c, \\ 0 & \text{if } r_{pw} > r_c. \end{cases} \quad (12)$$

A plot of the function $\Psi(r_{pw})$ with $\kappa = 0.1$ as shown in Fig. 4. In all our simulations, we use this value henceforth.

The collision of fluid particle with a wall is assumed to be elastic and specular as shown in Fig. 5(c). The specular reflection is implemented as follows [24],

$$\hat{\mathbf{d}}_s = 2(\hat{\mathbf{d}}_n \cdot \hat{\mathbf{d}}_i) \hat{\mathbf{d}}_n - \hat{\mathbf{d}}_i, \quad (13)$$

where $\hat{\mathbf{d}}_i$ denotes the incident direction from the surface, $\hat{\mathbf{d}}_n$ the unit normal from the point of reflection and $\hat{\mathbf{d}}_s$ the unit vector along the direction of specularly reflected vector.

The classical no-slip is obtained in all the previous studies by reintroducing the fluid particles that escape through the wall, back into the domain with a negative velocity. It has been reported that this unphysical change in direction influences the temperature near the wall especially in the direction tangential to the wall [21]. We impose no-slip by increasing the lateral dissipation (friction) between the fluid particles and its corresponding IFP. This is executed by increasing the dissipative force tangential to the wall on the fluid particle nearer to the boundary by means of increasing the lateral friction coefficient γ_{pw}^S (refer Eqs. (8) and (9)) greater than γ_{pp}^S . This increase in transverse dissipation forces reduces the relative velocity of the particles in the direction tangential to the wall. Thus this method resorts to changing physical interaction parameters and does not rely on velocities and positions of the reintroduced particles to control the slip.

The lateral dissipation forces between a fluid particle and the wall are assumed to be greater than the fluid–fluid interaction forces. Due to this larger lateral dissipation forces, the relative velocity component of fluid particles tangential to the wall surface is reduced. We assume the lateral dissipation coefficient is a function of distance from wall r_{pw} and it varies as

$$\gamma_{pw}^S = \begin{cases} \alpha(1 - r_{pw}/r_c)^2 \gamma_{pp}^S, & \text{if } r_{pw} < r_c, \\ 0 & \text{if } r_{pw} > r_c, \end{cases} \quad (14)$$

where α is a slip modification parameter. The chosen form of γ_{pw}^S ensures that, the lateral friction between fluid and solid particles increase as the fluid particle moves nearer to the wall. The dissipation and random coefficients are related through $\sigma_{pw}^S = \sqrt{2k_B T \gamma_{pw}^S}$ to avoid temperature fluctuations due to the modified drag coefficient. The translational, rotational and ran-

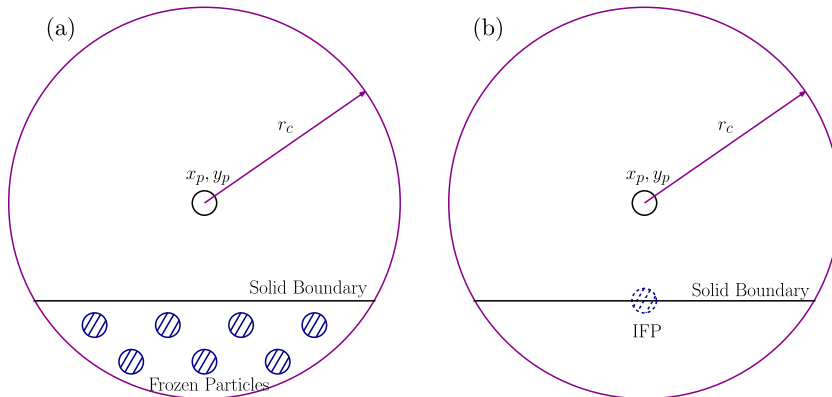


Fig. 3. A comparison of interaction between a fluid particle at (x_p, y_p) and the solid wall for (a) FWB and (b) IWB schemes.

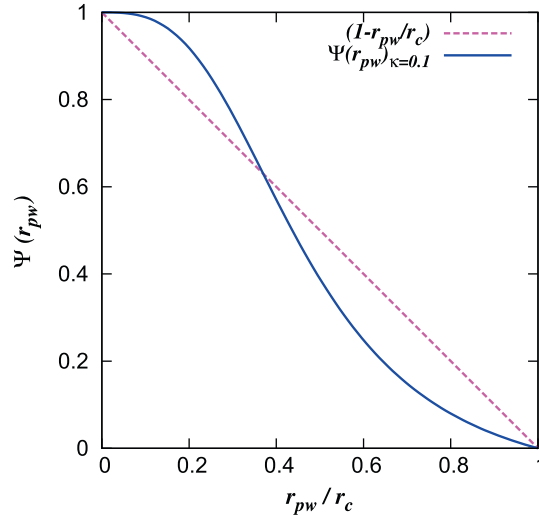


Fig. 4. Plot of the particle wall conservative weight function $\Psi(r_{pw})$ given in Eq. (12).

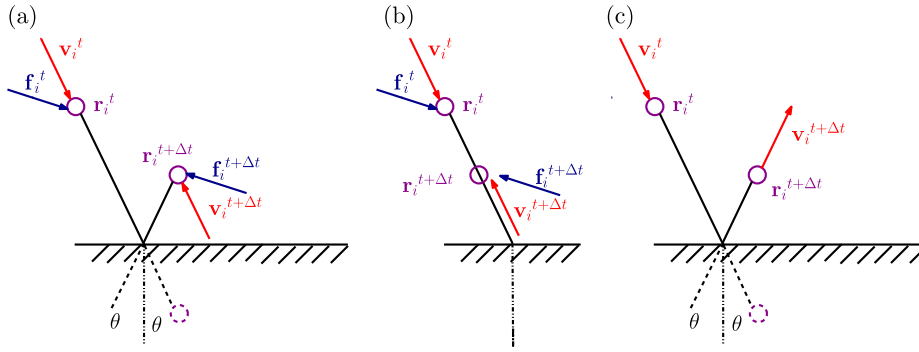


Fig. 5. A comparison of reinsertion schemes for fluid particles traversing through the solid wall: (a) bounce forward reflection [32], and (b) bounce back reflection [32], and (c) specular reflection. In the present work collision of a fluid particle with the wall is considered to be elastic and specular.

dom force components of fluid–wall interaction is computed using Eqs. (8)–(10) replacing γ_{ij}^s and σ_{ij}^s with γ_{pw}^s and σ_{pw}^s respectively and the rest of the central components of forces remain the same. The fluid particles near the wall experience large resistance to tangential relative motion due to the increased lateral forces and eventually the particles near to the wall attain the velocity of wall. The scheme is easy to implement and the partial slip boundary can also be easily obtained by changing the parameter α .

4. Simulation of benchmark problems

In this section, a detailed investigation of three classical benchmark problems using the new stick-wall approach to model the solid boundary is presented, and results compared well with those from available literature. The cut-off radius chosen to compute the short range interactions between particles $r_c = 1$. In all the simulations that follow, the parameters are chosen such that the temperature of the system is $k_B T = 1$. The dissipation parameter is chosen to be $\gamma_{pp}^c = \gamma_{pp}^s = \gamma_{pw}^c = 4.5$, and the coefficient of the stochastic force $\sigma_{pp}^c = \sigma_{pp}^s = \sigma_{pw}^c = 3$. All the fluid and boundary particles are assumed to be of the same size so that the size parameter is $\lambda_{ij} = \lambda_{ji} = \frac{1}{2}$. To reduce the computational time, all particles in each unit cell are linked through a cell list [25]. The forces on each particle are calculated considering only the neighboring cells so that the computational complexity of algorithm is $O(N)$. The temporal evolution of particle positions is obtained by integrating the momentum Eqs. (1) and (2) using the modified velocity-Verlet computational scheme of Groot and Warren [11] with a time step of $\Delta t = 0.01$. The temperature is calculated using the fluctuation of the particle velocities around the mean velocity. The translational and rotational kinetic energy were monitored separately and the temperature associated with each component is

$$(k_B T)_{Trans} = \sum_i \frac{m_i \langle v_i^2 \rangle}{d}, \quad (k_B T)_{Rot} = \sum_i \frac{I_i \langle \omega_i^2 \rangle}{d}, \quad (15)$$

where $d = 2$ for 2D simulations.

4.1. Planar Poiseuille flow

We first consider steady flow between two infinite parallel plates in two dimensions using FPD and the new IWB to model solid walls. The physical dimensions of the planar domain is chosen to be $30r_c \times 30r_c$. Each unit cell of dimension $r_c \times r_c$ is filled with FPD particles of uniform size at random positions initially with number densities of $\rho = 3, 6, 9$ or 12 particles per unit cell.

The simulation box is periodic in the x -direction (the flow direction) so that a particle crossing right or left boundary reappears on the opposite end with the same velocity. The maximum repulsion parameter a_{ij} is calculated using $a_{ij} = 75k_B T / \rho$ for different number densities following [11].

The fluid which is initially at rest is driven by a steady external force $\mathbf{f}_{ij}^{\text{External}} = 0.02$ applied on every particle in x -direction. The variation of the translational and rotational temperature with respect to time are plotted in Fig. 6(a). The temperature of the fluid is zero initially and it increases with time. The dissipative and random components of the inter-particle force act as a thermostat to ensure that the temperature achieves a steady state value of $k_B T = 1$. The net momentum of the particles in the simulation box increases due to the externally applied force and is, eventually, balanced by the dissipative forces at steady state as seen in Fig. 6(b).

The density, velocity, and temperature are obtained by averaging over 10^5 time steps after reaching the steady state. The channel is subdivided in y -direction into 300 bins to obtain the spatial distribution of density and velocity while 30 bins were used to obtain the spatial distribution of temperature.

We perform simulations to determine the repulsion parameter for fluid–solid interaction (a_{pw}) such that the density fluctuations near the solid boundary are minimum. Similarly, we determine the slip modification factor of transverse dissipation coefficient α , to obtain no-slip condition at the wall. As can be seen in Fig. 7, the value of α affects the amount of slip. The values of the repulsion parameter and the slip modification parameter for transverse dissipation coefficient depend on the density of particles and we list the values in Table 1 for various values of the particle density. We note that tunable slip can be achieved by adjusting the factor α . Our simulations indicate that the choice of α does not depend on the shear rate for $Re < 1200$.

The velocity, density and temperature profiles were compared against theoretical values. The velocity in x -direction (u) from FPD simulation is normalized with respect to net velocity (u_{average}) averaged over 10^5 iterations after reaching equilibrium. The velocity profile is plotted with the analytical solution for velocity of flow between two infinite parallel plates given by

$$\frac{u}{u_{\text{average}}} = 1.5 \left(1 - \frac{y^2}{h^2} \right),$$

where y is the distance from center line to the wall and $h = 15$ is the half width of channel. As can be seen from Fig. 8 there is excellent agreement of simulation results with analytical result. A density plot across the channel is shown in Fig. 8 and percentage maximum density fluctuation is calculated as 8.33% and also observed that most of the density variations are within the wall cut-off radius. The maximum density fluctuation ratio is obtained by dividing the change in maximum and mean system densities $\Delta\rho$ within the 300 horizontal bins by system density ρ .

The simulated velocity field matches well with the analytical profile for several other values of the density of particles as well as plotted in Figs. 9–11. The maximum density fluctuation ratios for these cases are 5.12% for $\rho = 6$ and 6.99%, for $\rho = 9$, and 8.58% for $\rho = 12$. For all densities, the near-wall velocity profiles also matches well with analytical solutions as shown in Fig. 12.

Pivkin and Karniadakis [22] have reported that the density fluctuation increases as the number density increases for FWB method. However, in our implementation we note that there is not much of variation in maximum density fluctuation ratio with the number density of particles. A comparison of maximum density fluctuation for different number densities for the FWB method and new IWB method is shown in Fig. 13. The spatial variation of both translational as well as rotational tem-

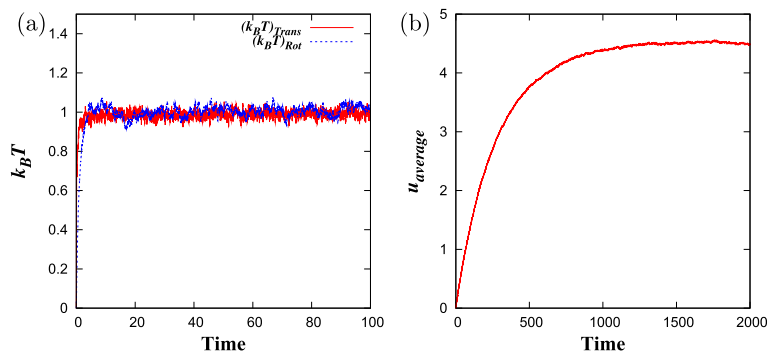


Fig. 6. Variation of (a) domain temperature and (b) average momentum with time for $\rho = 3$.

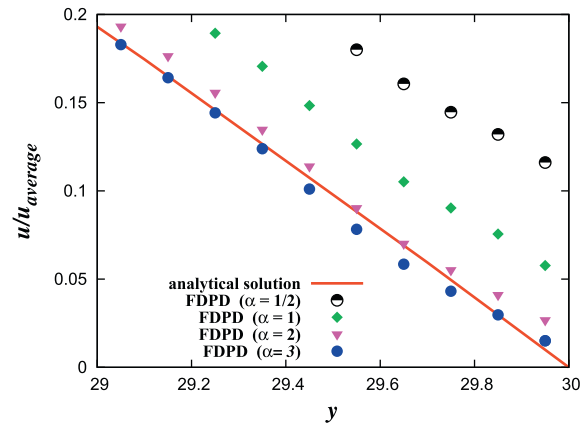


Fig. 7. Velocity profile within cut-off distance from the wall, for different slip modification parameters $\alpha = \frac{1}{2}, 1, 2$ and 3 for number density $\rho = 3$. The slip length decreases with increasing α .

Table 1

Simulation parameters for planar Poiseuille flow with varying number density.

ρ	a_{pp}	a_{pw}	α
3	25.0	20.00	3.0
6	12.5	21.00	4.5
9	8.33	21.25	5.0
12	6.25	21.5	6.5

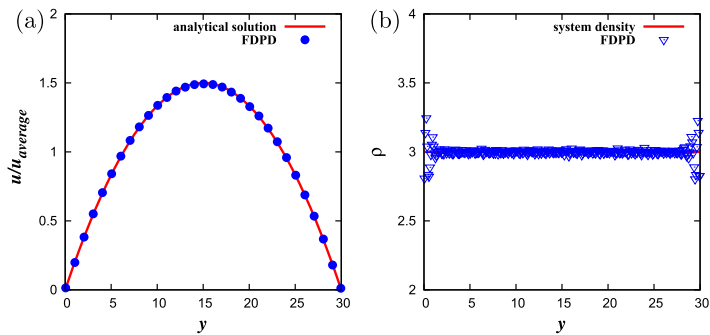


Fig. 8. (a) Velocity and (b) density distribution across the channel for the fluid particle number density $\rho = 3$.

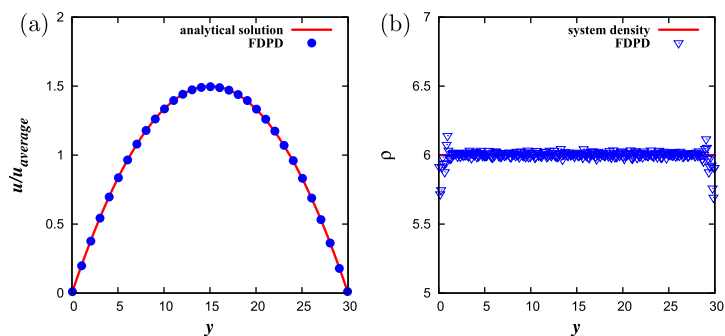


Fig. 9. (a) Velocity and (b) density distribution across the channel for the fluid particle number density $\rho = 6$.

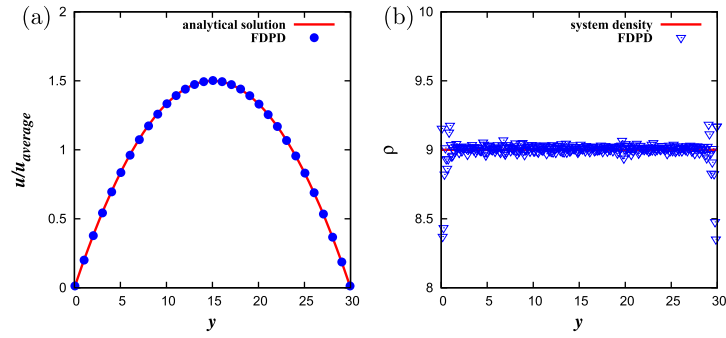


Fig. 10. (a) Velocity and (b) density distribution across the channel for the fluid particle number density $\rho = 9$.

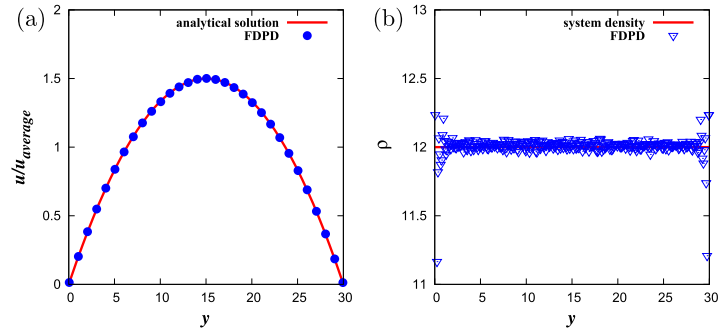


Fig. 11. (a) Velocity and (b) density distribution across the channel for the fluid particle number density $\rho = 12$.

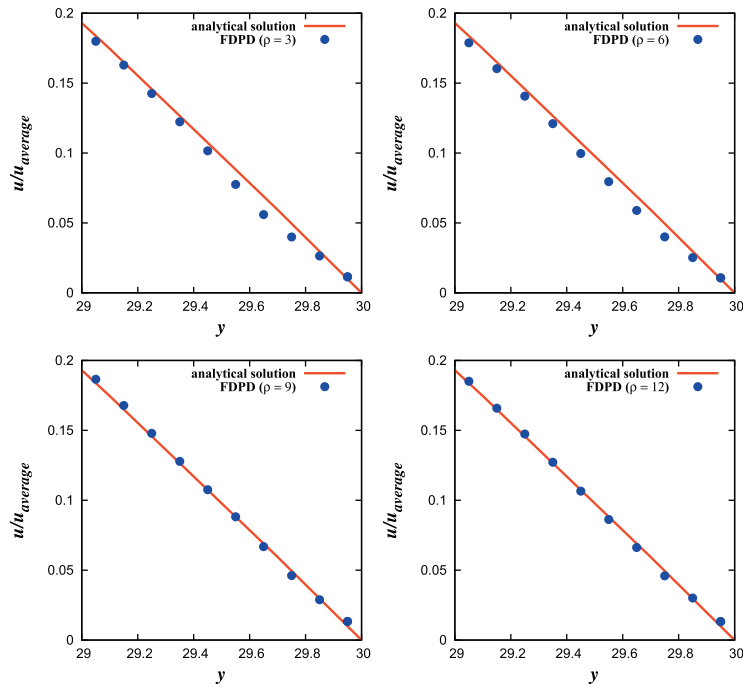


Fig. 12. Enlarged view of velocity distribution near the solid wall within cut-off radius r_c , for different densities $\rho = 3, 6, 9$ and 12 .

peratures are shown in Fig. 14 and it can be seen that the temperature profiles are not significantly affected by the presence of the wall.

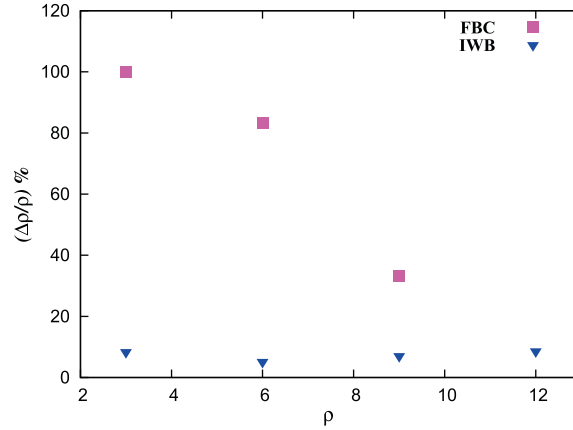


Fig. 13. Comparison of the maximum density fluctuation (extracted from [7]) in FBC method and IWB method, for different densities.

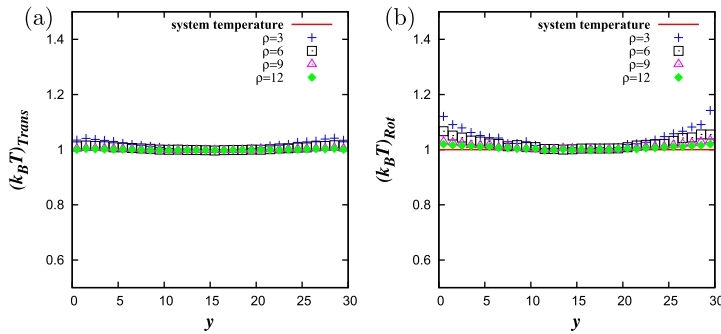


Fig. 14. Spatial variation of temperature across the channel for density ($\rho = 3, 6, 9$ and 12), (a) translational temperature and (b) rotational temperature.

4.1.1. Viscosity of FDPD fluid

The dynamic viscosity (μ) can be calculated using kinetic theory [26] and is given by the expression [9]

$$\mu = \frac{45(k_B T)^2}{2\pi\sigma^2 r_c^3} + \frac{\pi\rho^2\sigma^2 r_c^5}{105(d^2 + 2d)k_B T}. \quad (16)$$

The first term in the above expression arises from the kinetic contribution and the second, from dissipative contributions. The viscosity obtained from DPD simulations generally varies from this theoretical prediction [9], and so several numerical viscometers have been devised to find the viscosity. Simulations using a periodic Poiseuille flow method (PPFM) as numerical viscometer Backer et al. [9] reported that viscosity of a fluid modeled by standard DPD is greater than viscosity calculated using Eq. (16). In the present work the viscosity of DPD fluid is determined using the method described by Fan et al. [2] from the planar Poiseuille flow simulations. A parabolic profile based on continuum theory is fitted for the velocity in x -direction (u) obtained from the simulation

$$u = u_{\max} \left(1 - \frac{y^2}{h^2} \right)$$

and the maximum velocity $u_{\max}$ is extracted. The viscosity is then calculated from the expression

$$u_{\max} = \frac{\rho g h^2}{2\mu}, \quad (17)$$

where μ is the dynamic viscosity of fluid.

The dynamic viscosity μ for FDPD fluids with different densities ρ is calculated with driving force $g = 0.02$ and half width $h = 15$ and are presented in Fig. 15. It is observed that the viscosity of the FDPD fluid is uniformly greater than viscosity obtained by Backer et al. [9] for densities ranging from 3 to 10. The additional rotational and lateral dissipative force components in FDPD scheme appear to make the FDPD fluid more viscous than the standard DPD fluid. A similar observation was made previously by Junghans et al. [27] while studying the transport properties of DPD fluids using TDPD scheme.

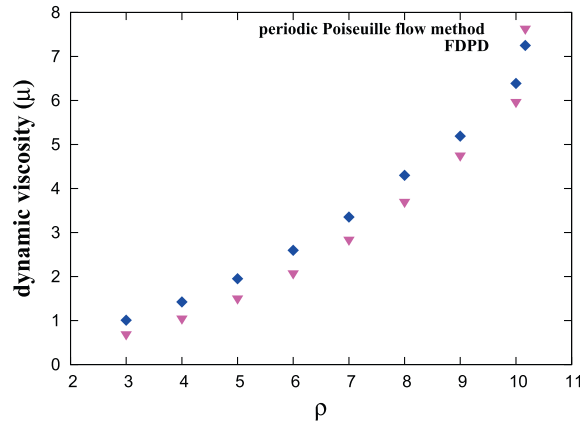


Fig. 15. A plot of viscosity of FPD and standard DPD fluids. The viscosity of standard DPD fluid is extracted from the PPFM results of Backer et al. [9].

4.1.2. Impenetrability of new wall (IWB)

The penetrability of fluid particles through the solid boundary was compared for a FWB method (standard DPD) and new IWB scheme for number density $\rho = 4$, by simulating a planar Poiseuille flow with a domain size of $30r_c \times 30r_c$ with $\Delta t = 0.01$. In case of the FWB method, the solid boundaries were modeled by fixing two layers of frozen particles following [18] and the number of particles traversing the boundary were counted for 10^5 iterations. It was observed that, the average number of particles escaping per wall ($30r_c$) per time step is 4.77 for FWB method and 0.42 for IWB method. For IWB method, on an average, less than one particle crosses the boundary per time step. This is because of the large conservative repulsion coefficient used in the present scheme and also due to the repulsion of fluid particles near the wall along normal direction. By the introduction of the new IWB scheme to model solid walls, we come closer to modeling a perfectly impenetrable no-slip wall with minimum density fluctuations for DPD simulations.

4.2. Lid driven cavity for low Reynolds number

We next study flow in a square driven cavity of dimensions $30r_c \times 30r_c$ shown in Fig. 16. The density of fluid particles is chosen to be $\rho = 6$ (the domain thus consists of 5400 fluid particles). The top lid is moved with uniform velocity U . The computational parameters used for the simulations are lateral friction coefficient $\alpha = 4.5$, sliding velocity $U = 1.44$ and repulsion parameter for wall $a_{pw} = 21$. The Reynolds number for the test case is calculated as $Re = \frac{\rho UL}{\mu} = 100$ where the dynamic viscosity, $\mu = 2.6$ for density $\rho = 6$. The time step for simulation $\Delta t = 0.01$ and the properties were averaged over 10^5 iterations as before. The velocities of particles in each unit cell (total 900 cells) are averaged to find the velocity vectors.

For comparison, we also solved the incompressible Navier–Stokes equations using an open-source CFD tool box OpenFOAM. A grid size of 30×30 is used to discretize a square cavity of $0.1 \text{ m} \times 0.1 \text{ m}$ dimensions and is solved using the incompressible flow solver of OpenFOAM with a time step of $\Delta t = 0.001$. The cavity is filled with a fluid having kinematic viscosity $\nu = 1 \times 10^{-3} \text{ m}^2/\text{s}$ and the upper lid is moved with velocity 1 m/s . The velocity vector plots from the FPD simulations and the CFD simulations are qualitatively very similar as shown in Fig. 17.

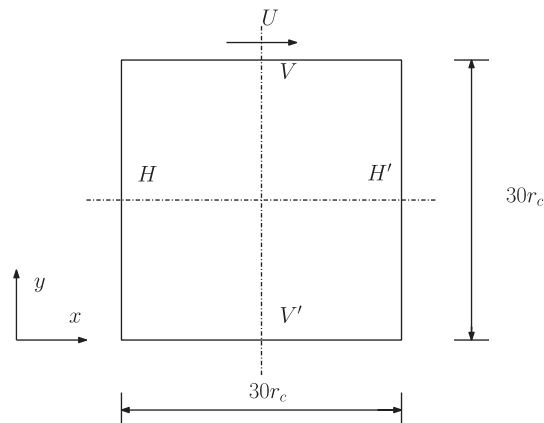


Fig. 16. Square cavity with top wall moving with velocity U in x -direction.

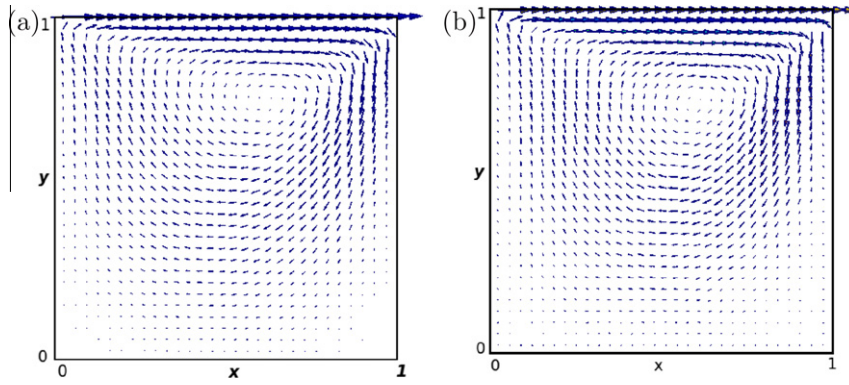


Fig. 17. Velocity vector plots of flow in a square cavity (upper lid moving) obtained from (a) OpenFOAM and (b) FPD simulations for $Re = 100$.

A quantitative comparison with the results of Ghia et al. [28] is done by extracting both velocities in x and y -directions from horizontal (HH') and vertical (VV') middle sections as shown in Fig. 18. The velocity profiles at these sections for $Re = 100$ is compared against the numerical prediction of Ghia et al. and also against Navier–Stokes solutions using OpenFOAM [29]. A good comparison can be noticed. The density profiles across the HH' and VV' sections are plotted in Fig. 19 and it can be seen that the density fluctuation ratios are 5% and 6.03%, respectively.

4.3. Flow past a circular cylinder

We next apply our method to another well known benchmark problem and study the flow past a circular cylinder. A cylinder of diameter D is placed at $x = 50r_c$ and $y = 75r_c$ in a square domain of dimensions $150r_c \times 150r_c$ with density $\rho = 3$. The total number of fluid particles in the simulation is 67,267 and fluid viscosity $\mu = 1.01$. A uniform flow of velocity U_∞ along x -direction is introduced at the left end of the channel as shown schematically in Fig. 20. Particles lying between $x = 0$ and $x = r_c$ in x -direction are supplied velocities $u = U_\infty$ and $v = 0$ throughout the simulation. The domain is ensured to be large enough to maintain the free stream fluid properties far away from the cylinder in both the x and y -directions. We set periodic boundary conditions in y -direction.

Simulations were carried out for different Reynolds numbers (defined as $Re = \frac{\rho U_\infty D}{\mu}$) in the range $Re = 10$ to 40. Table 2 lists cylinder diameters and uniform flow velocities chosen. To avoid compressibility effects, the Mach number (ratio of flow velocity to isothermal speed of sound $c_s = 3.05$) is set below 0.33. The wall repulsion parameter was chosen to be $a_{pw} = 20$ and slip modification parameter $\alpha = 1.5$.

The results were ensured to be domain size independent by calculating the drag coefficient on a larger domain size $200r_c \times 200r_c$ containing 119,767 particles at the maximum Reynolds number 40. It was observed that the difference in drag coefficient is negligible for both cases so all the simulations reported here use a domain size of $150r_c \times 150r_c$. For a simulation involving 1.5×10^5 iterations, using an IntelTM i5 (3.6GHz) processor, about 6 h of CPU time were required.

All the fluid particles in a layer of thickness r_c around the cylinder feel the presence of the solid wall. The inter-particle force between the fluid particle and the wall acts along the line connecting fluid particle and center of cylinder. Particles are reflected from the cylinder surface by specular non-dissipative reflection using the Eq. (13). The reflection of the particles occurs at the point of intersection of its trajectory with the surface of circle. In most of previous work using other schemes

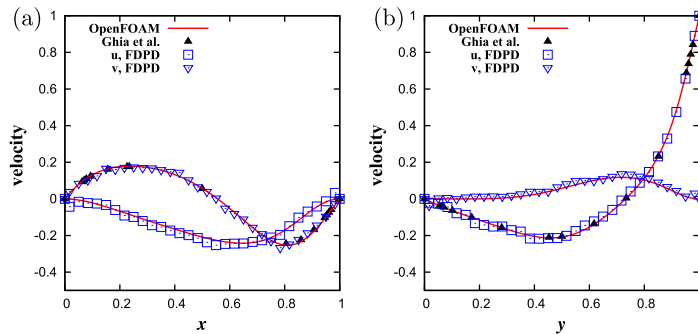


Fig. 18. Comparison of velocity profiles (u, v) in a lid-driven cavity, across (a) the horizontal mid section at $y = L/2$ (HH'), and (b) vertical midsection at $x = L/2$ (VV') for $Re = 100$.

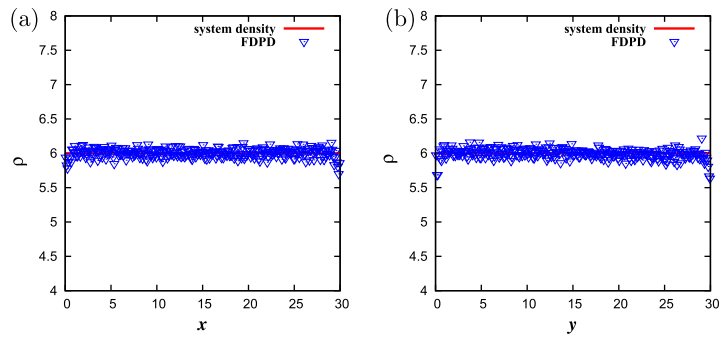


Fig. 19. Mid-sectional density variation in a square cavity along (a) HH' and (b) VV' for $\rho = 6$.

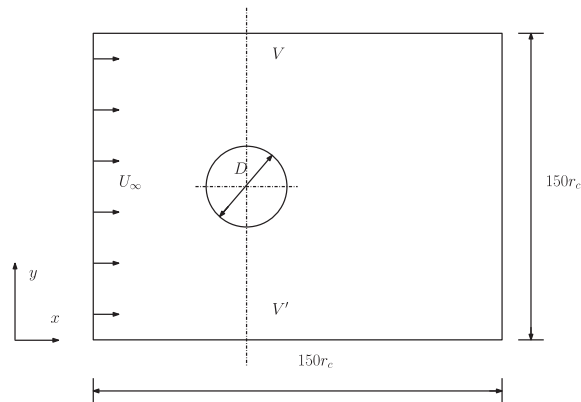


Fig. 20. Circular cylinder in a uniform flow with velocity U_∞ in the x -direction.

Table 2

Reynolds and Mach numbers for different cylinder diameters and uniform flow velocities.

D	U_∞	Re	$Ma\left(\frac{U_\infty}{c_s}\right)$
14	0.96	40	0.31
10	1.01	30	0.33
10	0.67	20	0.22
10	0.33	10	0.11

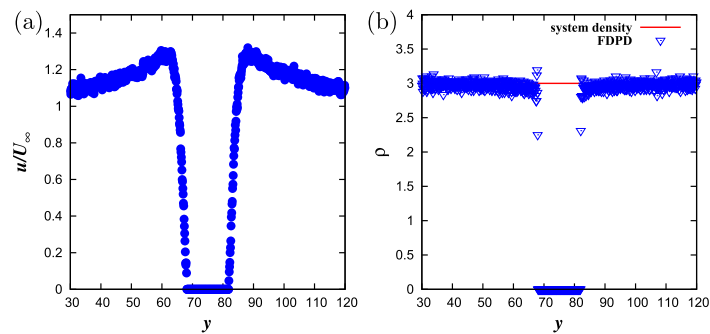


Fig. 21. (a) Velocity and (b) density profiles across the vertical section VV' for $Re = 40$.

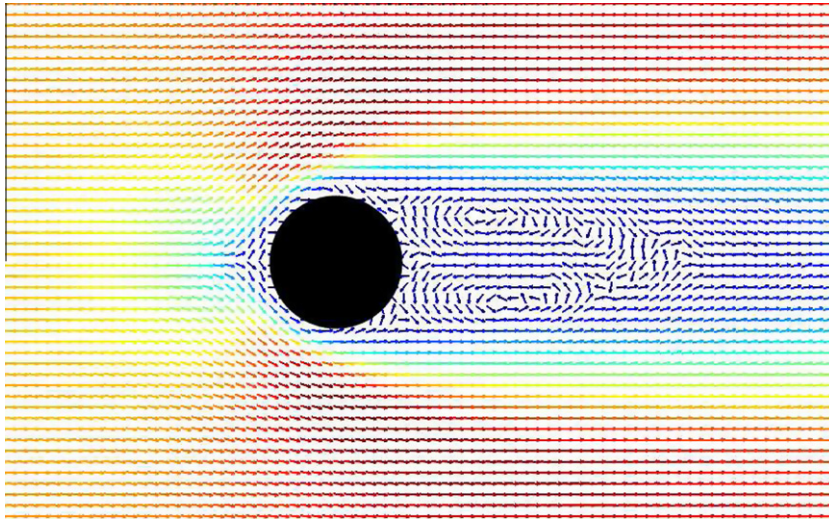


Fig. 22. Velocity vector plot of flow past circular cylinder for $Re = 40$.

Table 3

Comparison of drag coefficients of circular cylinder in uniform flow for different Reynolds numbers.

	$Re = 10$	$Re = 20$	$Re = 30$	$Re = 40$
Tritton [30]	2.93	2.08	1.76	1.58
Altenhoff et al. [31]	2.82 ± 0.11	2.08 ± 0.03	1.82 ± 0.02	1.69 ± 0.02
De Palma et al. [19]	3.18	1.95	–	–
Present study	2.99	2.14	1.74	1.66

for boundary conditions, the tangential component of the velocity is *reversed* to obtain the no-slip boundary condition. In our scheme, since no-slip is achieved due to the lateral dissipation force between the particle and solid wall, we do not have to resort to such an unphysical velocity reversal.

The radial density and velocity profiles were extracted along vertical cross section VV' and plotted in Fig. 21. We note that the density is uniform throughout the cross-section and no-slip can be observed from the velocity profiles. Due to the presence of the bluff body, the flow accelerates near the solid wall and the free stream conditions were recovered far from the cylinder; $\frac{u}{u_\infty}$ at the both upper and lower periodic boundaries was found to be less than 1.05. Two recirculating eddies known as Föppl vortices are visible behind the cylinder for $Re = 40$ as shown in Fig. 22.

For a quantitative comparison, the drag coefficient was calculated by computing the force acting on the cylinder in the direction of flow. The drag force is obtained by the vector sum of the forces exerted by the surrounding particles within r_c radially from the cylinder surface on the wall averaged over 10^5 iterations [10]. A comparison of drag force obtained from the current simulations with earlier experimental work [30] and DPD simulations [19,31] is given in Table 3, and shows good agreement.

5. Summary

In this work we developed and implemented a new no-slip boundary condition for a modified DPD scheme which is envisaged to simulate non-homogeneous (with different size) rotating and translating particles. In this model linear and angular momentum balance equations are solved simultaneously to update positions, linear and angular velocities of the particles. Instantaneous wall particles are used to mimic a continuous solid boundary with an effective infinite wall particle density. We show that, this model of a continuous solid wall in combination with modified repulsion potential for the central forces between fluid particles and the wall can be used to minimize spurious fluctuations in the fluid density and temperature near the wall. Moreover, with this approach the walls are naturally impenetrable. No-slip boundary conditions are obtained by modifying the lateral dissipative force components of fluid particles approaching the wall. Internal (Poiseuille and lid-driven cavity) and external flows (flow past circular cylinder) were simulated using the new scheme and the results matched analytical, experimental, and numerical results to good accord. This model can be easily extended to three dimensional simulations.

The new scheme addresses the disadvantages of solid wall modeling with fixed particle methods such as FWB and SWB. The computational time for the newly proposed scheme is lesser than the other frozen particle methods due to fewer fluid–

solid interactions. Another feature of this scheme is that the slip is *tunable* by controlling the lateral dissipation parameter α and it is possible to achieve perfect slip, partial slip or no-slip as required by the physics of the problem. Density oscillations due to the presence of a solid wall is well controlled; not exceeding 8.58% for all densities for planar walls. The proposed scheme is easy to implement for more complicated wall geometries. This scheme is also applicable to the DPD models in which particle rotation is not considered, such as the one presented by Filipovic et al. [13].

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